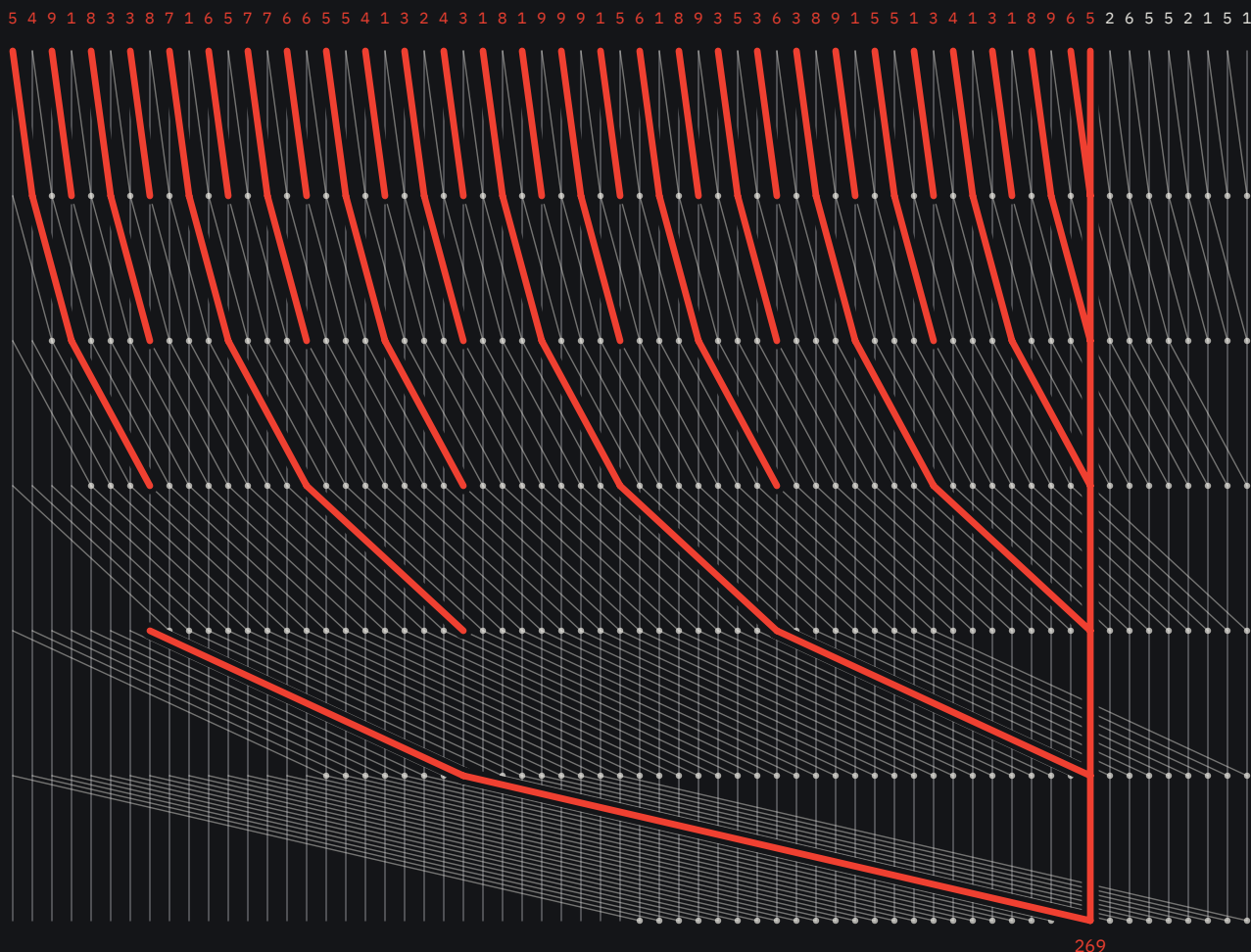


# Kernel Engineering

*From silicon to speculative decoding – GPU kernels for modern LLMs.*



**FIG. 0** A Kogge-Stone prefix sum, the pattern inside GPU scan kernels: 64 threads, 6 steps, 321 additions. At step  $k$  each thread adds the partial sum  $2^k$  places to its left. In red: the additions thread 55 depends on, gathering 56 of the digits above into 269.